Reliability on EUV Interconnect Technology for 7nm and beyond

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Abstract— In this research, we present the robust reliability performance of BEOL by utilizing EUV single patterning and further feasibility of EUV process for future nodes. As compared to ArF, EUV BEOL shows superior reliability performances such as significantly improved TDDB, breakdown voltage (Vbd), robust resistance shift with SM and TC tests and reliable package level characteristics. Long-term TDDB analysis follows an even more aggressive power law model rather than the root E model under low bias. Newly developed via array (quasi-power rail) EM structure has a great feasibility to enhance EM performance for future nodes.

Keywords- BEOL, EUV, Reliability, Via array EM structure.

I. Introduction

Moore's law scaling trend projects the dimension of interconnect to scale by approximately 0.7X every technology generation. On-chip interconnect (BEOL) scaling has become one of the most critical challenges for recent CMOS technology nodes. In place of multi-patterning, multi ArF immersion lithography processes and respective masks are required to form the desired image, an extreme ultra violet (EUV) provides more advantageous options in terms of cost and pattern shapes to use single-mask exposure. In response to aggressive scaling challenges, Samsung 7 nm FinFET technology utilizing EUV introduces a new metallization solution for the 7 nm node and further advanced nodes to meet the reliability and performance requirement at the local interconnect layers for Electromigration (EM) enablement. More details on the EUV solution used for this study can be found in References [1, 2].

When scaling BEOL dimensions, reliability requirements such as time-dependent dielectric breakdown (TDDB) and EM have to meet. Although fine-line dimension scale with each node, operating voltages and current densities do not scale proportionally, which causes the burden on reliability. In this study, we demonstrate the reliability characterization of BEOL for 7 nm FinFET technology that was newly developed by utilizing EUV and further feasibility of EUV process beyond 7 nm node.

II. Experimental

The hardware used for this study was fabricated with 7 nm FinFET technology that was newly developed by utilizing EUV. Cu lines with Cu seed and Ta/TaN/Co liner were embedded in dense ULK dielectric and capped with Co layer. EM tests were carried out under constant current at various temperature range (300~340 C°). IMD TDDB test was performed under constant voltage on both EUV single patterning and ArF immersion multi patterning. Voltage ramping test at 0.1V/0.03s was also conducted for the direct comparison of breakdown voltage between ArF and EUV. Stress migration (SM) and wafer level Temperature cycling (TC) tests with various structures were performed in the various temperature ranges. SM stress temperature was 175 C°, 200 C°, 225 C° and 250 C° and wafer level TC stress condition was from -60 C° to 150 C°.

III. Results and Discussion

The interconnect stack is constituted of 14 interconnect layers with ULK dielectrics as shown Fig. 1 (a). EUV was straightforwardly used for single patterning of MOL and BEOL, not just as a means for cutting of self-aligned quadruple patterning (SAQP) and self-aligned double patterning (SADP). The minimum pitch metal is 36nm at M1. As compared to ArF patterning, the adoption of EUV single patterning for MOL and BEOL process shows significant improvement of 2D fidelity as well as uniformity, suggesting strong immunity to weak points in random layout (Fig. 1 (b)).

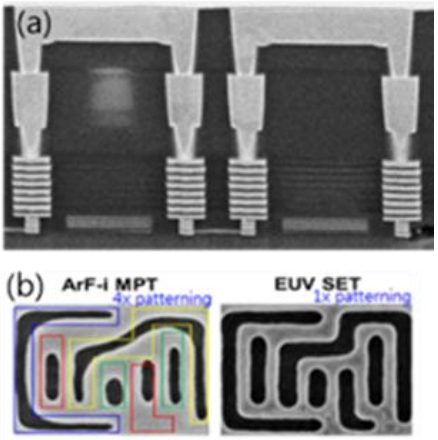

Figure 1. (a) Representative cross sectional TEM image of BEOL by utilizing EUV single patterning. (b) EUV 2D fidelity is around 70% superior to ArF.

Fig. 2 shows direct comparison of via-to-metal TDDB distribution results fabricated by EUV single patterning and ArF, respectively. Inset exhibits the schematic diagram of Vx-Mx TDDB test structure. As shown in Fig. 2, via-to-metal TDDB distribution results using EUV single patterning exhibits significantly improved uniformity, resulting from strong immunity of EUV to weak points in random layout, especially the weakest point in via to under metal spacing. It is noteworthy that TDDB Vmax of 7nm EUV showed approximately 1.1 times higher than that of 10nm ArF.

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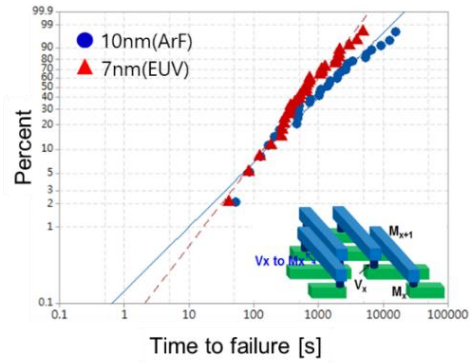


Figure 2. Vx-Mx TDDB distributions for 10nm (ArF) and 7 nm (EUV) process. Inset shows the schematic diagram of Vx-Mx TDDB test structure.

As displayed in Fig. 3, although ArF also shows the substantially stable Vbd, direct comparison of Vbd distribution also shows that EUV has more tight distribution than ArF. Note that more than 0.7V gain which translates in more than 1.8 nm space gain was achieved at 1 ppm.

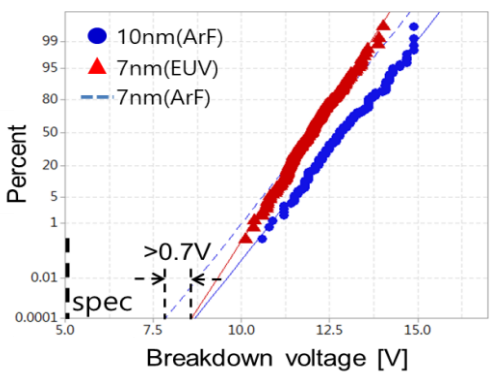
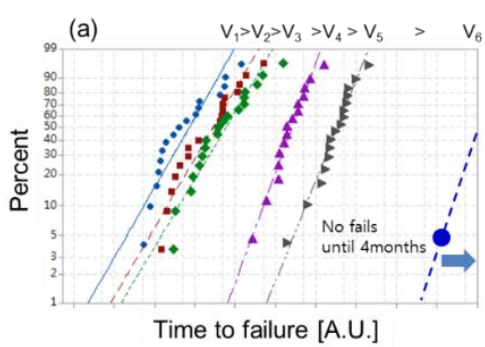


Figure 3. Breakdown voltage between 10 nm (ArF) and 7 nm (EUV). Dotted line indicates shift by 10nm vs. 7nm spacing difference.

As exhibited in Fig. 4, long-term TDDB experiments were performed on Vx-Mx structure having minimum design spacing. Similar trend to previous study [3], the tighter distribution trend was observed with no failures up to 4 months for stress in the lowest stress field (Fig. 4 (a)). However, such a change in Weibull slope might be regarded as similar considering the slope of low percentile in the high voltage stress distributions. Under high bias, the current conduction follows Poole-Frenkel (PF) or Schottky and thus follows the root E model [4, 5]. However, it follows an even more aggressive power law model rather than the root E model (Fig. 4 (b)).



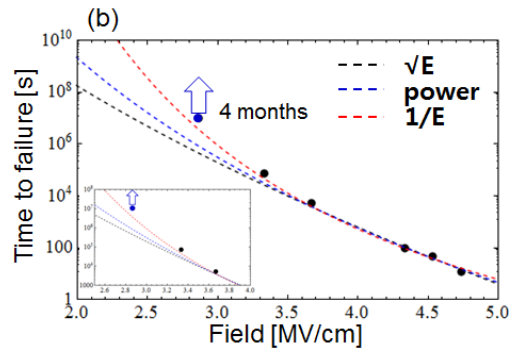


Figure 4. (a) TDDB distribution for each stress voltage which is including wafer and package level long-term stress results. (b) t63.2% vs. stress field trend for each stress field.

Fig. 5 (a) and (b) show leakage current-stress and gamma E (Field acceleration factor) extracted from TDDB experiments with various spacing respectively. As shown in Fig. 5 (a), hard breakdown (HBD) is clearly observed at high bias (> 5.5V) and charge injection follows root E model. However, soft breakdown (SBD) is observed at low fields (<5.5 V) because electrons have insufficient energy to overcome energy barrier thus TDDB degradation deviates from root E model [6]. Gamma E nearly shows constant in a wide range of spacing (> 18 nm) spacing. However, as the spacing reduces below 18 nm, this relation no longer holds and expected to change from field driven to voltage driven model (Fig. 5 (b)). These different signatures observed on different voltages and spacing indicate various breakdown mechanisms under different stressing conditions. Therefore, the aggressive TDDB lifetime model, i.e. power law or 1/E, should be considered for future technology nodes.

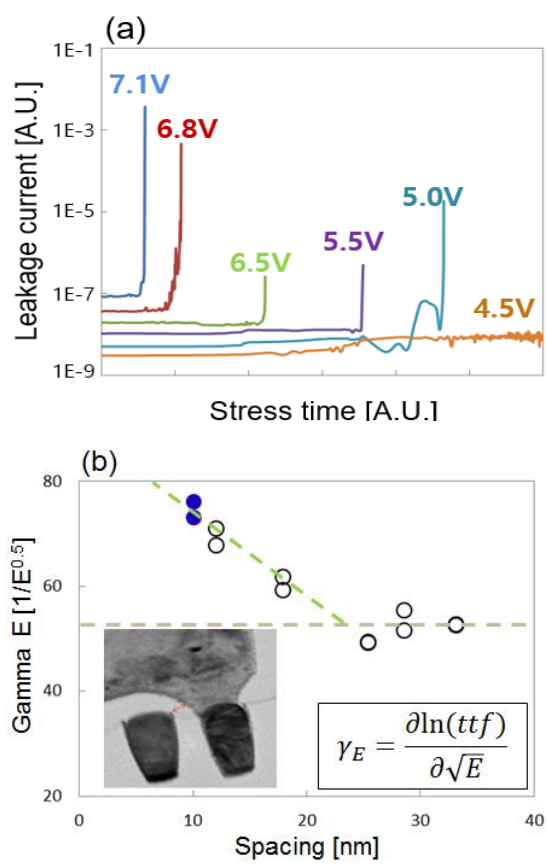


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Figure 5. (a) Leakage current vs. stress time for each stress voltage. Soft breakdown was showed at low stress voltage level (5V) (b) gamma E (Field acceleration factor) vs. spacing extracted from TDDB experiments including prior nodes (blue circle : 7nm EUV).

As shown in Fig. 6, EM testing was performed on both "electron up" and "electron down" configurations. Those test structures evaluated in this work were illustrated in insets of Fig. 6. Negligible difference was observed on both EUV and ArF, however, more tight EM distributions (σ: 0.5 → 0.4) were obtained on both configurations with respect to that of ArF through process optimization such as barrier metal, electroplating, etc.

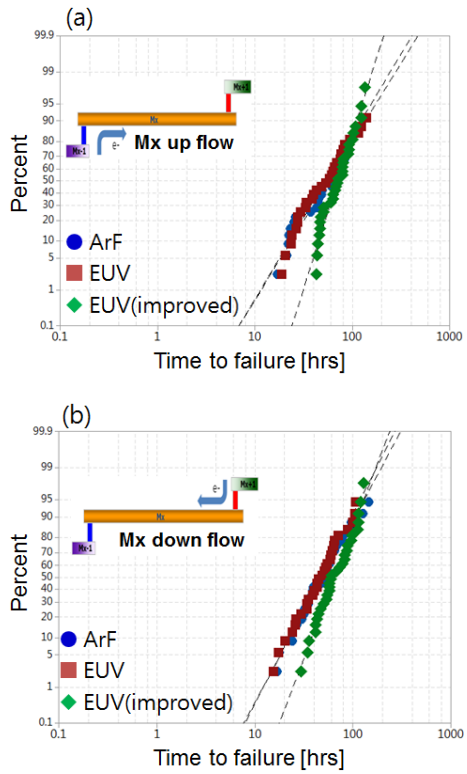

Figure 6. (a) "electron up" and (b) "electron down" flow EM Log-Normal distributions with process optimization respectively. Insets show the schematics of "electron up" and "electron down" EM structures.

Fig. 7 shows the results of SM and wafer level TC tests with various structures in the various temperature ranges. As can be seen in Fig. 7, negligible resistance shift was observed on both SM and TC tests, indicating robust BEOL structures utilized by EUV.

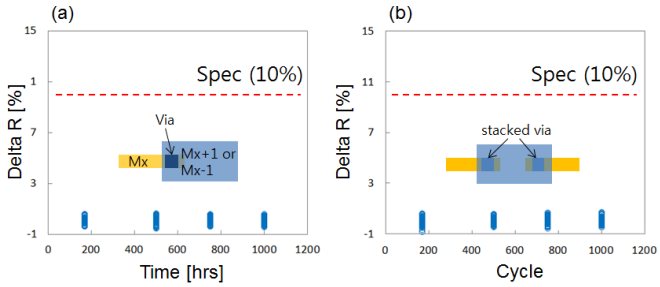

Figure 7. Wafer level SM delta R versus time plot for one of the worst structure (Iso-via with wide metal). (b) TC delta R versus cycle plot for stacked via structure. Insets show the schematics of SM and TC structures.

In response to aggressive scaling challenges, we newly developed via array EM structure (quasi-power rail) to enhance EM performance for future node, as shown in Fig. 8 (a). EM log-normal distribution demonstrated that newly designed via array EM test structure (upper via structure) showed significant EM performance uplift, approximately 2X higher, and robust resistance shift as compared to conventional single connected EM test structure (Fig. 8 (b)). Those improvements are contributed to the suppression of Cu void growth rate (ΔR/Δt), resulting from increased number of Cu/TaN interfaces working as metallic bonding (Fig. 8 (c)). In addition, in case of lower level via array structure, no noticeable difference was observed unlike upper via structure, indicating negligible reservoir effect on this structure. However, this reservoir can suppress back stress effect, resulting in reduction of short length gain [7].

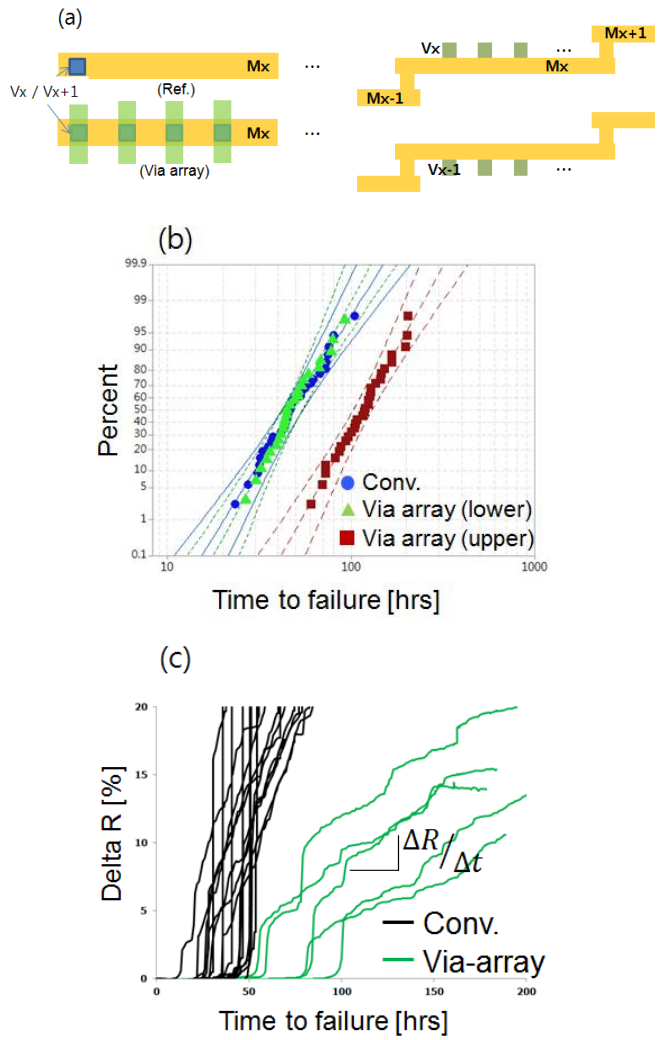

Figure 8. (a) Schematic diagram of newly designed EM macro structures (Conv. vs. Via array structure). Right shows cross sectional schematic of via array EM structure. (b) EM log-normal distribution of conventional single via and via-array structures. (Failure criteria : 10% delta R) (c) Time versus delta R plot for both conventional and via-array structure during EM test.

CPI (chip package interaction) reliability evaluations were performed to prove the thermo-mechanical properties. Test structures of CPI TV were electrically tested utilizing various chains to inspect the under bump crack, corner delamination and leakage current. Package level reliability tests for CPI follow JEDEC standards (J-STD 020D, JESD22-A104D,

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JESD22-A118, JESD22- A103). We obtained substantially robust reliability results with any failures as summarized in Table 1.

Table 1. Summary of CPI Qualification results

Stress item		Condition	Duration	A-1	A-2	A-3
HTS		150℃	1000 hours	Pass	Pass	Pass
Precon (MSL3)		Bake 125℃ Soak 30℃/60% RH Reflow 260℃	24 hours 192 hours 3 cycles	Pass	Pass	Pass
	TC	B(-55℃~125℃)	≥1000 cycles	Pass	Pass	Pass
	u-HAST	130℃, 85% RH	96 hours	Pass	Pass	Pass

IV. Conclusions

In this study, we demonstrate the reliability characterization of BEOL for 7 nm FinFET technology that utilized the EUV lithography. The results also showed superior reliability characteristics that could enable further feasibility of EUV process beyond 7 nm node. EUV single patterning for BEOL process shows significantly improved reliability properties including CPI such as superior TDDB and Vbd distribution, negligible resistance and leakage current with SM and TC as compared to ArF. Long-term stressed TDDB followed even more aggressive power law model rather than root E model. Additionally, it is noteworthy that newly developed via array (quasi-power rail) EM structure has a great feasibility to enhance EM performance for future node.

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